

TWO NON-BOLD WHEEL ENTRIES OF PARIDA–MOURA’S TABLE 4 ARE EQUALITIES:

$$t(W_{11}) = t(W_{12}) = 8$$

ABSTRACT. For a graph G on n vertices let $t(G)$ be the least t admitting a G -cover-free family on the ground set $[t]$, in the sense of Parida and Moura. Table 4 of their preprint tabulates $t(P_n)$, $t(C_n)$, $t(W_n)$ and $t(K_n)$ for $n \leq 12$ under the caption “Upper bounds . . . where the bold entries indicate exact values”; the two wheel entries $t(W_{11}) \leq 8$ and $t(W_{12}) \leq 8$ are printed non-bold. We show that both are equalities,

$$t(W_{11}) = t(W_{12}) = 8.$$

Table 4 leaves the exactness of these two non-bold upper bounds undetermined, and no statement of that preprint asks whether they are exact; what we do is supply the two matching lower bounds and so turn those two printed bounds into equalities. The upper bounds come from two families exhibited in full below. The lower bounds are exhaustive non-existence statements: there is no W_{11} -CFF(7, 11) and no W_{12} -CFF(7, 12), established by an exhaustive search over the whole solution space.

1. THE DEFINITION AND THE TWO TABLE ENTRIES

Let $G = ([1, n], E)$ be a graph. Following Parida and Moura [1, Definitions 3.1 and 4.1], a family $(B_1, \dots, B_n)$ of subsets of $[1, t]$, with B_i sitting on vertex i , is a *G -cover-free family* G -CFF(t, n) when

- (1) $B_a \setminus B_b \neq \emptyset$ and $B_b \setminus B_a \neq \emptyset$ $(\{a, b\} \in E)$,
- (2) $B_{i_0} \setminus (B_a \cup B_b) \neq \emptyset$ $(\{a, b\} \in E, i_0 \notin \{a, b\})$,

and

$$t(G) = \min\{t : \text{a } G\text{-CFF}(t, |V(G)|) \text{ exists}\}.$$

Throughout, P_n and C_n are the path and the cycle on n vertices and W_n is the wheel on n vertices: a hub joined to every vertex of a rim cycle C_{n-1} . This is the convention of [1], where W_n is a universal vertex joined to all vertices of a cycle of length $n - 1$.

The two entries at issue are in Table 4 of [1], whose columns are n , $t(P_n)$, $t(C_n)$, $t(W_n)$, $t(K_n)$. In the $t(W_n)$ column that table lists a bold 7 for $n = 10$ and a plain 8 for each of $n = 11$ and $n = 12$, under the caption “Upper bounds for $t(P_n)$, $t(C_n)$, $t(W_n)$, and $t(K_n)$, where the bold entries indicate exact values.” The table is introduced with the words “we display

2020 *Mathematics Subject Classification.* 05D05, 05B40, 05C90.

Key words and phrases. cover-free family, cover-free family on a graph, wheel, Sperner family, exhaustive search.

the known values for $t(P_n)$, $t(C_n)$, and $t(W_n)$ or their upper bounds for $n \leq 12$ ". So by the source's own caption these two plain entries stand there as upper bounds whose exactness is left undetermined; we determine them below. No statement of [1] asks whether they are exact: the source contains no conjecture, and its five Open Problem statements concern other objects.

Theorem 1. $t(W_{11}) = 8$ and $t(W_{12}) = 8$.

Section 2 proves $\leq$ by exhibiting the two families; Section 3 proves $\geq$ by an exhaustive census.

2. THE UPPER BOUND: TWO EXHIBITED FAMILIES

The upper bound 8 is already printed in [1]. The two paragraphs below are elementary and are recorded only so that the two witnesses exhibited here are self-contained; no novelty is claimed for them. We use the following observations about (1)–(2).

Lemma 2. *Let $m \geq 3$ and let $(B_1, \dots, B_m)$ be a C_m -CFF(t, m), with the cycle $1, 2, \dots, m, 1$. Then $B_i \setminus B_j \neq \emptyset$ for all $i \neq j$; in particular the B_i are nonempty and pairwise distinct.*

Proof. Fix $i \neq j$ and read indices modulo m . If $i \in \{j-1, j+1\}$ then $\{i, j\}$ is an edge and (1) gives $B_i \setminus B_j \neq \emptyset$. Otherwise $i \notin \{j, j+1\}$, so (2) applies to the edge $\{j, j+1\}$ with $i_0 = i$ and yields $\emptyset \neq B_i \setminus (B_j \cup B_{j+1}) \subseteq B_i \setminus B_j$. $\square$

Lemma 3. *Let $m \geq 3$, let $(B_1, \dots, B_m)$ be a C_m -CFF(t, m) on $[1, t]$, and set $B_0 = \{t+1\}$. Then $(B_0, B_1, \dots, B_m)$ is a W_{m+1} -CFF($t+1, m+1$), where the hub is vertex 0 and the rim is the cycle $1, 2, \dots, m, 1$. Consequently $t(W_{m+1}) \leq t(C_m) + 1$.*

Proof. Every B_i with $i \geq 1$ lies in $[1, t]$ and $B_0 = \{t+1\}$ is disjoint from all of them; all sets lie in $[1, t+1]$. There are two kinds of edge.

Hub edges $\{0, i\}$. For (1): $B_0 \setminus B_i = \{t+1\} \neq \emptyset$, and $B_i \setminus B_0 = B_i \neq \emptyset$ by Lemma 2. For (2) with $i_0 = k \notin \{0, i\}$: $B_k \setminus (B_0 \cup B_i) = B_k \setminus B_i \neq \emptyset$, again by Lemma 2.

Rim edges $\{i, i+1\}$. Condition (1) is inherited from the C_m -CFF. For (2) with $i_0 = 0$: $B_0 \setminus (B_i \cup B_{i+1}) = \{t+1\} \neq \emptyset$; and with $i_0 = k \geq 1$, $k \notin \{i, i+1\}$, it is inherited from the C_m -CFF as well. $\square$

Lemma 4. *A G -CFF(t, n) is also a G -CFF($t+1, n$). Hence if no G -CFF(t, n) exists then $t(G) \geq t+1$.*

Proof. Neither (1) nor (2) mentions the ground set; enlarging $[1, t]$ to $[1, t+1]$ leaves every set, every difference and every union unchanged, and $B_i \subseteq [1, t] \subseteq [1, t+1]$. The second sentence follows because the set of t admitting a G -CFF(t, n) is therefore an up-set. $\square$

Here are the two objects. On the ground set $\{1, \dots, 7\}$, indexed along the rim cycle $v_1, v_2, \dots$, put

$$(3) \quad \begin{aligned} R_{10} = & (\{1\}, \{2, 3\}, \{2, 4\}, \{2, 5\}, \{2, 6\}, \\ & \{6, 7\}, \{3, 6\}, \{3, 5\}, \{5, 7\}, \{4, 7\}), \end{aligned}$$

$$(4) \quad \begin{aligned} R_{11} = & (\{1\}, \{2, 3, 4\}, \{2, 3, 5\}, \{2, 3, 6\}, \{2, 7\}, \{2, 4, 6\}, \\ & \{4, 5, 6\}, \{3, 4, 6\}, \{3, 4, 7\}, \{3, 5, 7\}, \{5, 6, 7\}). \end{aligned}$$

Lemma 5. R_{10} is a C_{10} -CFF(7, 10) and R_{11} is a C_{11} -CFF(7, 11).

Proof. Direct inspection of (1)–(2). For R_{10} this is 10 edges, the last of them the wrap edge $\{v_{10}, v_1\}$, hence 20 instances of (1) and $10 \cdot 8 = 80$ instances of (2); for R_{11} it is 11 edges, 22 instances of (1) and $11 \cdot 9 = 99$ of (2). Every one is a set difference between explicitly listed sets of size at most 3. The accompanying program performs all 221 of them and reports the number of violations, which is 0 in both cases. $\square$

Proof of $t(W_{11}) \leq 8$ and $t(W_{12}) \leq 8$. Apply Lemma 3 with $t = 7$ to R_{10} and to R_{11} , which are C_{10} - and C_{11} -cover-free families by Lemma 5. With $B_0 = \{8\}$ this gives, on the ground set $\{1, \dots, 8\}$, a W_{11} -CFF(8, 11)

$$(\{8\}; R_{10})$$

and a W_{12} -CFF(8, 12)

$$(\{8\}; R_{11}),$$

the hub carrying $\{8\}$ and the rim carrying (3) respectively (4) in the printed order. $\square$

3. THE LOWER BOUND: AN EXHAUSTIVE CENSUS

Proof of $t(W_{11}) \geq 8$ and $t(W_{12}) \geq 8$. By Lemma 4 it suffices to show that no W_{11} -CFF(7, 11) and no W_{12} -CFF(7, 12) exists. Each is a finite question: an assignment of a subset of $[1, 7]$ to each of n vertices, so $(2^7)^n$ tuples, namely 2^{77} for $n = 11$. It is decided as follows.

Symmetry. Conditions (1) and (2) are built from set differences and unions only, so the whole constraint system is invariant under the symmetric group S_7 acting on the ground set. The orbits of S_7 on subsets of $[1, 7]$ are exactly the size classes, so a solution exists if and only if one exists in which the set on the first vertex of the search order is one of the eight nested representatives $\emptyset, \{1\}, \{1, 2\}, \dots, \{1, \dots, 7\}$. Restricting the first vertex to those eight masks is therefore exact, not a heuristic.

Decomposition. Fixing in addition the set on the second vertex of the search order, arbitrarily among all $2^7 = 128$ subsets, splits that canonical space into $8 \times 128 = 1024$ shards. They are pairwise disjoint, and their union is the whole canonical space, because every canonical tuple has exactly one pair of first two sets.

Enumeration. Each shard is enumerated by depth-first search over the remaining vertices, taking all 2^7 subsets at each, and backtracking on (1)

or (2) as soon as every vertex a condition mentions has been assigned. No derived lemma is used as a prune, so an empty census is a statement about (1)–(2) alone.

Outcome. All 1024 of 1024 expected shards return, and the totals are

	solutions	shards	nodes
W_{11} at $t = 7$:	0	1024/1024	14,002,112
W_{12} at $t = 7$:	0	1024/1024	14,015,792

By Lemma 4, $t(W_{11}) \geq 8$ and $t(W_{12}) \geq 8$. □

Theorem 1 follows by combining the two sections.

4. WHAT WAS CHECKED

The accompanying program `verify.py`, whose recorded run is `verify.output.txt`, checks the two printed witnesses (3)–(4) and their hub extensions directly against (1)–(2) and re-runs the complete $t = 7$ censuses for W_{11} and W_{12} , reproducing the shard, solution and node counts reported above.

The upper bound is hand-checkable from (3)–(4) with 221 elementary set differences; the lower bound rests on two exhaustive censuses, which the program re-runs rather than replaces by a hand proof. Only the two equalities of Theorem 1 are established: the other non-bold cells of Table 4 are neither proved nor used, no $t(W_n)$ with $n \geq 13$ is claimed, and no least-order claim is made — nothing here says W_{11} is the smallest wheel whose table entry was left inexact. The transcription and interpretation of Table 4 are read off the e-print of [1] rather than produced by the computation.

REFERENCES

- [1] P. Parida and L. Moura, *Cover-free families on graphs*, arXiv:2605.12634v1 [math.CO], 12 May 2026. arXiv:2605.12634. Definition, table and numbering references are to v1.